

Full live demo walkthrough (click-by-click)

Read this straight through, live, in front of the professor. Every step names the exact thing to type or click, what should appear on screen, and a short line to say out loud. **Total spoken time target: ~9:30 for the core demo** (see the timing table at the end for the full breakdown; padding included for questions). The core demo (sections 1-7, 13, 14) runs entirely inside the GUI: two chat windows plus a live **Security Dashboard**, with a real relay attack triggered by clicking a button in the **Attack Lab** panel and caught live by the other window's own checks. Sections 8-12 (Wireshark and the standalone `demo/run_*.py` scripts) are optional padding / a backup if any live click doesn't cooperate.

Assumes: Windows or any OS with Python 3.8+, this repo cloned, and `pip install -r requirements.txt` already run. If not, run that first — it's not counted in the demo time.

1. Pre-demo setup (before the professor arrives — not part of the timed demo)

Do this once, quietly, before anyone is watching.

Fastest way (macOS): double-click `demo/start_demo.command` in Finder (or run `demo/start_demo.sh` in a terminal). It does 1.1–1.3 below **and** step 2 for you: it stops any leftover server of ours, checks the certificate (regenerating it only if it is missing or invalid), opens a titled Terminal window for the server (`--lab`), waits until the server is accepting connections, then opens two chat windows and the security dashboard. Check the server window shows the line in 1.2. Use `--no-lab` for normal mode or `--port N` for another port; stop everything with `demo/stop_demo.sh`. If you use it, skip to step 3 (the two login windows are already open). The manual steps follow, for reference or if you are not on a Mac.

1.1 Open a terminal in the project root and run the environment reset script — **recommended even if you think certs already exist**, since you'll likely be on a fresh or different machine for the actual presentation, where `certs/` won't exist at all yet:

```
python demo/reset_environment.py
```

This reports (and asks before killing) a leftover `server.py` still listening on the server's port — it never touches other programs, such as macOS's AirPlay Receiver, which also uses port 5000 — then (re)generates `certs/server.crt` / `server.key` and prints the fingerprint of the cert it just wrote. **Note that fingerprint down** — you'll confirm it matches the server's own startup line in the next step. (If you'd rather generate certs manually instead: `python certs/generate_certs.py`, skippable if `certs/server.crt` and `certs/server.key` already exist from a previous run on this machine.)

1.2 In that same terminal, start the relay server:

```
python server/server.py --lab
```

`--lab` turns on the opt-in **Attack Lab**: it has no effect at all unless a `lab_control` request also comes from an already-logged-in, localhost connection (see ["How this differs from WhatsApp/Signal"](#) in the main README), so this is safe to leave on for the whole demo.

Confirm it printed a line like this before moving on (the port will read `5000` unless you passed `--port`, and the fingerprint will be whatever `reset_environment.py` just printed — **these two must match**; if they don't, see ["Troubleshooting: certificate mismatch"](#) in the main README):

```
[*] Server listening on 127.0.0.1:5000 over TLS (cert fingerprint: XXXX XXXX XXXX XXXX) (routes handshake/chat envelopes; never sees plaintext app secrets or session keys)
```

Leave this terminal running and visible for the whole demo — minimize it to a corner of the screen; you'll point at it in step 5 to show it never prints anyone's message content.

1.3 In a second terminal, open the security dashboard (it only reads `logs/security_events.jsonl` — it never connects to the server, so it's harmless to leave open for the whole demo, including right now before anything has happened):

```
python gui/security_dashboard.py
```

Drag it to a corner of the screen, visible but out of the way of the two chat windows you'll open next.

1.4 Quick sanity check (optional but recommended): run `pytest tests/ -v` once and confirm it ends with `337 passed`. This is not shown to the professor; it's your own confidence check that nothing is broken before you start.

2. Launch two GUI clients — ~30 sec

If you used `demo/start_demo.command` in step 1, the two chat windows and the dashboard are already open — arranged top-left, top-right and bottom-left; click a window's title bar to bring it forward — so skip to step 3.

2.1 Open a second terminal, at the project root, and run:

```
python gui/chat_gui.py
```

A dark-themed window titled "**Secure Chat (GUI)**" appears, showing a login card labeled "Secure Chat" with fields for Server host, Server port, Username, Peer username, and Password. Drag this window to the **left half** of the screen.

2.2 Open a third terminal and run the exact same command again:

```
python gui/chat_gui.py
```

A second, identical window appears. Drag this one to the **right half** of the screen, so both are visible side by side, with the server terminal from step 1.2 visible somewhere below or behind them.

Say: "This is the same client logic as our terminal version — the GUI just wraps it, so what you're about to see is the real protocol, not a mockup."

3. Register user 1 (left window) — ~45 sec

3.1 In the **left** window: click the **Server host** field — it's pre-filled with `127.0.0.1`, leave it as-is. Click the **Server port** field — pre-filled with `5000`, leave it as-is.

3.2 Click the **Username** field and type: `alice`

3.3 Click the **Peer username** field and type: `bob`

3.4 Click the **Password** field and type any password, e.g. `demo-pass-1234` (it will show as dots as you type).

3.5 Click the **Register** button.

Expected on screen: the login card is replaced by the chat screen. At the top you see `alice ↔ bob`, and the **Conversation** panel shows a centered gray line reading `Your key fingerprint: XXXX XXXX XXXX XXXX` (16 hex characters in 4 groups). The **Wire Log** panel on the right starts filling with lines like `--> register: username=alice, ...` and `<-- register_result: success=True, ...`.

Say: "Registering just created a fresh RSA-2048 identity for alice — the private key never leaves this machine; only the public key went to the server, and you can see that exact envelope in the wire log on the right."

4. Register user 2 (right window) — ~45 sec

4.1 In the **right** window, repeat exactly: click **Username**, type `bob`; click **Peer username**, type `alice`; click **Password**, type any password (can be different from alice's — they're independent accounts), e.g. `demo-pass-5678`.

4.2 Click **Register**.

Expected on screen: within a couple of seconds, **both** windows' Conversation panels show a centered line: `Secure session established with <other username>`. The top-right of each window now shows a highlighted strip: `"s fingerprint: XXXX XXXX XXXX XXXX"`. The Wire Log on both sides shows `handshake_init / handshake_response` lines with truncated `pubkey=...` and `sig=...` values.

Say: "The moment both of us were online, the two clients automatically ran an ECDH handshake — signed with our RSA keys so the server can't substitute its own key in the middle, which we'll prove directly in demo 3 later."

5. Compare fingerprints out loud — ~45 sec

5.1 Point at the highlighted fingerprint strip in the **left** (alice's) window, top-right corner. Read the 16 hex characters out loud, e.g. "seven, four, six, B..."

5.2 Point at the same strip in the **right** (bob's) window and read it too.

Expected: the fingerprint alice's window shows for "bob" is **character-for- character identical** to the fingerprint bob's own window shows for himself as "Your key fingerprint" earlier — and vice versa.

Say: "If a relay server — or anyone — tried to slip in a different key mid- handshake, these two fingerprints wouldn't match, and we'd catch it just by reading them aloud to each other, independent of any code running. This is the exact human backstop our design doc calls 'check key fingerprints.'"

Messages to type (doctor ↔ pharmacist script)

Use these exact lines so the audience sees a realistic hospital exchange. **Window A = Dr. Rao (the doctor, e.g. user `dr-rao` , Peer `pharmacy`) ; Window B = the pharmacist (user `pharmacy` , Peer `dr-rao`).** Each line is short enough to read on a projector. Type it in the window shown, press Enter. (Any two usernames work; `dr-rao` sorts before `pharmacy` , so window A starts the handshake, which is what the MITM step needs.)

Normal conversation (step 6):

#	Type in	Message
1	A (doctor)	Patient 4471: amoxicillin 500 mg, 3 times daily for 7 days.
2	B (pharmacist)	Received. Checking the allergy record first.
3	B (pharmacist)	No allergies on file. Dispensing now.
4	A (doctor)	Thanks. Please log the dispensing time.
5	B (pharmacist)	Logged at 14:32. Ready at counter 2.

Attacks (step 7). Arm in the doctor's window (A), then type the message shown in A:

Attack	Arm (button in A's Attack Lab)	Then type in A	What B shows
Tamper	Tamper next message	Correction: dose is 1000 mg, not 500 mg.	red "Message blocked: tampered in transit" card
Replay	first send a normal line: Patient 4471 is also due a blood test. , wait for it to appear in B, then Replay last message	(nothing: the relay resends it)	"Message blocked: replayed message"
Drop	Drop next message	Cancel the 14:00 delivery. then immediately Please confirm the new time.	amber "Warning: message missing" on the second line, which is still shown
MITM	Before B logs in: arm MITM next handshake in A, then log in B (restart B's window if it is already logged in; the other order never triggers it)	(nothing)	red "Handshake blocked: signature check failed" and sending disabled

What to expect from the hash chain: after a **tamper**, the next normal message shows an amber "1 message missing ... matches a message blocked earlier" card (the tampered line really was sent, so its number is skipped; it is reported once, and later messages are green again). After the **drop**, the second line reports "1 message missing ... possible deletion by the relay" and messages after it are verified normally. Each attack is reported exactly once.

After the attacks: B (pharmacist): `Understood. Confirmed for 15:30.` then click **Export Evidence** in either window for the signed record.

6. Exchange messages both directions — ~90 sec

6.1 In the **left** (alice) window, click the message entry field at the bottom and type: `hi bob, this is alice`. Click the **Send** button (or press Enter).

Expected: a teal, right-aligned bubble with that text and a timestamp appears in alice's own Conversation panel. In bob's (right) window, a dark-gray, left-aligned bubble with the same text appears. In **both** Wire Log panels, a new line appears: `--> chat: from=alice nonce=... ciphertext=... tag=... (alice's side) / <-- chat: from=alice nonce=... ciphertext=... tag=... (bob's side)` — truncated base64, never the plaintext.

6.2 In the **right** (bob) window, type: `hey alice, got it` and click **Send**.

Expected: mirrored — teal bubble on bob's side, gray bubble on alice's side, chat envelope lines in both wire logs.

6.3 Send one more each way (any text) so there are at least 2–3 message bubbles each direction visible before moving on.

6.4 In the **right** (bob) window, **click one of alice's gray bubbles**. A small dark popup titled **Security receipt** opens. Read out: the sender and their fingerprint, **AES-GCM tag ✓ verified**, **RSA signature ✓ verified**, **Timestamp ✓ fresh (N s old)**, **Chain position: seq #1 ✓ links to the previous message**, and the truncated nonce and record SHA-256. Press Esc or **Close**. Then click one of bob's own teal bubbles to show the sent version ("✓ signed with your private key").

Say: "Every message carries its own receipt: not 'trust us, it's encrypted', but exactly which checks this specific message passed. Everything on it was verified by the client before the message was ever displayed."

Say: "Notice the Conversation panel reads like an ordinary chat app — that's the point. Now look at the Wire Log: every message we just read in plain English exists on the network only as this — nonce, ciphertext, tag. Nobody watching this connection, including our own relay server, can read what we just said to each other."

7. Attack Lab: live relay attacks from the GUI — ~3:30 min

Both windows show an amber **Attack Lab** button in the top bar (it's only enabled because the server is running with `--lab`). Click it in the **left** (alice) window to open the panel — point at the amber **"LAB MODE: relay is acting maliciously on request"** banner first.

Say: "Everything in this panel is a real attack, carried out by our actual relay server against its own connection, not a simulation. What catches it is the exact same client-side code we've already shown — nothing special runs when lab mode is on."

7.1 Tamper. Click **Tamper next message**. The result line reads `Armed: next chat message from alice will be tampered with`. Type a message (e.g. `transfer approved`) and click **Send**. **Expected:** alice's window shows the result line change to `Performed by relay: flipped a byte in...`; the **right** (bob) window shows a red system notice `REJECTED message from alice: AES-GCM authentication failed (tampered or wrong key)` — the tampered text itself is never displayed. The dashboard's event feed gains a red `lab_attack_performed` line and a red `security_alert` line (bob's own client reporting what it just caught).

Say: "The relay really did flip a bit of the ciphertext — and our AES-GCM tag catches it, same as the standalone tamper demo, just triggered from inside the app."

7.2 Drop. Click **Drop next message**. Send a message — it **never appears** in bob's window at all (the relay silently discarded it; alice's own window still shows her own sent bubble, since that's drawn locally). Send one more message: bob's window now shows a red notice **"N message(s) missing ... possible deletion by the relay"** — the hash chain exposing the hole.

Say: "This is the one none of the other checks can catch on their own — a message that's dropped has nothing to fail authentication on, because it never arrives. The hash chain is what notices something's missing."

7.3 Replay. Send one ordinary message first (so there's something to replay), then click **Replay last message**. **Expected:** it fires immediately (not on your next send) — bob's window briefly shows the same message content arrive, followed at once by a red **"duplicate message detected ... rejected as a replay"** notice; the message is never displayed twice.

7.4 MITM next handshake (*timing-sensitive — see the note below*). This one substitutes the relay's own ECDH key into the handshake itself, so it only works **before** a session is already established. Close both windows, restart them, and in the **left** (alice) window — register but do **not** yet log in

from the right window — open Attack Lab and click **MITM next handshake** right away. Then register in the **right** (bob) window. **Expected:** bob's window shows `HANDSHAKE ABORTED` and never reaches "Secure session established"; sending a message from either side afterward re-attempts the handshake for real (now unarmed) and succeeds normally.

If the timing doesn't line up live, say so plainly and run the guaranteed-reliable version instead: `python demo/run_mitm_handshake_demo.py` (section 10 below) proves the identical defense deterministically.

7.5 Point at the dashboard's **event counts** row across all of the above — note the `lab_attack_performed` and `security_alert` counters climbing — and say:

Say: "Every one of those was a real attack and a real, independent catch, logged centrally without either chat window having to trust the other."

8. (Optional) Open the Wireshark capture for a static packet view — ~60 sec

Skip this step if short on time; it's a supplement to the live demo, not required.

Fresh capture instead (optional): `python demo/run_wireshark_demo.py --open-wireshark` generates a brand-new capture and opens it. It runs a scripted alice/bob session on a **spare port** with a **temporary data directory** (your real `data/` is not touched), then prints what it found. Expect: `Client Hellos: 2 TLS 1.3 Server Hellos: 2`, about 36 `Encrypted Application Data` records, and `None` of them appear anywhere in the `capture bytes` after searching the raw capture for the usernames, password, message text and JSON field names. With `--open-wireshark` the port is already decoded as TLS and the filter is `tls`; if you open the file by hand, use the port it prints: **Decode As... → TCP port → TLS**, then filter `tls`. Output file: `demo/wireshark_capture.pcapng` (scratch, overwritten each run). Details in [README.md](#). The steps below use the committed capture, which works identically.

7.1 Open Wireshark. **File → Open...**, navigate to and select `demo/capture_normal_session.pcapng`, click **Open**.

7.2 In the display filter bar at the top, type `tls` and press Enter to show only TLS-layer packets.

If the **Protocol** column shows `RSL` instead of `TLS`, Wireshark is guessing from the port number (5000 is officially a GSM "RSL" port). Right-click any packet → **Decode As...** → set **Current** to `TLS` → **Save** → **OK**. Do this once before the demo; Wireshark remembers it.

The capture holds two connections (alice's and bob's), each starting with its own TLS handshake.

7.3 Click the first packet whose **Info** column reads `Client Hello (SNI=localhost)`. In the packet-detail pane below, expand **Transport Layer Security** → **Handshake Protocol: Client Hello** to show the TLS handshake beginning the connection, and point at **Extension: server_name** → `localhost`: the server name the client is asking for.

7.4 Click the next packet, whose **Info** column starts `Server Hello`. Expand **Transport Layer Security** → ... → **Server Hello** and point at **Extension: supported_versions** → `TLS 1.3` and **Cipher Suite: TLS_AES_256_GCM_SHA384**.

Say: "The server's certificate is in this packet too, but TLS 1.3 encrypts it, so even the certificate is hidden from an observer. Everything after this Server Hello is encrypted."

7.5 Right-click any packet after the handshake completes (**Info** column reads `Application Data`) → **Follow** → **TCP Stream**. Point at the resulting hex/ASCII pane.

Expected: the reassembled stream is unreadable binary — no `{"type": "login"...}`, no readable text anywhere. To prove it, close that window, type `frame contains "alice"` into the display filter and press Enter: **zero packets** match, even though alice registered and chatted in this session. The same is true for `frame contains "password"` and for any of the message text.

Say: "This is a static capture of exactly the kind of session we just ran live. Every one of these 'Application Data' packets is one of our JSON envelopes — but from the outside, it's indistinguishable from noise. That's TLS doing its job on top of everything else we've already shown."

9. Backup: standalone tamper-detection script — ~60 sec, optional

Skip this whole backup section (9-12) unless something in the live Attack Lab (section 7) didn't cooperate, or you have extra time and want to show the exact same defenses proven a second, fully deterministic way.

8.1 In a fourth terminal (or reuse the one from step 1.3), at the project root, run:

```
python demo/run_tamper_demo.py
```

8.2 Let it run to completion (a few seconds). **Expected final output:**

```
=====
RESULTS
=====
[PASS] Normal message displayed correctly.
[PASS] Confirmed the malicious relay actually flipped a byte (this wasn't a no-op).
[PASS] Tampered message content was NEVER displayed to bob.
[PASS] Bob's client printed a clear tamper-detection warning.

=====
DEMO 1 PASSED
=====
```

Above that, point at the line reading `[!] WARNING: message from ... failed authentication (tampered or wrong key) -- discarded.`

Say: "This script starts its own throwaway server that deliberately flips one bit of a real message's ciphertext in transit — simulating a compromised relay. AES-GCM's authentication tag catches it immediately; the tampered content is never shown, only this warning."

10. Backup: standalone replay-protection script — ~60 sec, optional

9.1 Run:

```
python demo/run_replay_demo.py
```

9.2 Expected final output:

```
=====
RESULTS
=====
[PASS] Confirmed the malicious relay actually double-sent the envelope.
[PASS] The message content was displayed exactly once (1 occurrence(s)), not twice.
[PASS] Bob's client printed a clear duplicate/replay rejection warning for the second delivery.

=====
DEMO 2 PASSED
=====
```

Point at the line above reading `[!] WARNING: duplicate message detected from ... (same nonce seen before) -- rejected as a replay -- discarded.`

Say: "Here the malicious relay captures one real, validly-signed message and resends the exact same packet a second time — a classic replay attack. Even though it's byte-for-byte identical to a message that was already accepted once, the timestamp-plus-nonce tracking on the receiving side catches the duplicate."

11. Backup: standalone MITM handshake script — ~75 sec, optional

10.1 Run:

```
python demo/run_mitm_handshake_demo.py
```

10.2 Expected final output:

```

=====
RESULTS
=====
[PASS] Confirmed the malicious relay actually attempted the key swap.
[PASS] Bob's client derived NO session key from the forged handshake -- the MITM attempt was rejected.
[PASS] Bob's client printed a clear handshake-abort warning naming the possible MITM.

[*] For comparison -- what a human reading fingerprints aloud would see:
    Real demo_alice_....'s RSA fingerprint (what bob actually has on file):
        <16 hex chars>
    Attacker's RSA fingerprint (...):
        <different 16 hex chars>
[PASS] These fingerprints differ -- a human comparing them out-of-band (voice call, in person) would also catch this
attacker, independent of any automated check.

=====
DEMO 3 PASSED
=====

```

Point at the `HANDSHAKE ABORTED` line above, and then at the two different fingerprints printed at the bottom.

Say: "This is the attack our fingerprint step in part 5 defends against, proven directly: the relay tries to substitute its own key during the handshake itself — the classic unauthenticated Diffie-Hellman man-in-the-middle. Because we sign the handshake with our RSA identities, the victim's client refuses to derive a session key at all. And even in a hypothetical where that check somehow failed, you can see the resulting fingerprint wouldn't match — the human check from part 5 is a real, independent backstop, not just decoration."

12. Backup: standalone dropped-message script — ~60 sec, optional

11.1 Run:

```
python demo/run_drop_demo.py
```

11.2 Expected output (middle of the run, then the result):

```

--- STEP 2: alice sends message #2 -- but the relay silently DROPS it ---
[MALICIOUS RELAY] silently dropped a valid chat message (it is never forwarded, and nobody is told).

--- STEP 3: alice sends message #3 -- delivered; bob's chain check fires ---
[!] WARNING: 1 message(s) missing before #3 (expected #2): possible deletion by the relay.
...
[PASS] Bob's client detected the gap: '1 message(s) missing ... possible deletion by the relay'.
...
DEMO 4 PASSED

```

Point at the `[MALICIOUS RELAY] silently dropped` line, then at the `WARNING: 1 message(s) missing` line two steps later.

Say: "Encryption and signatures prove a message is real, but they can't tell you about a message that never arrives. Here a compromised relay silently drops a 'STOP, allergic reaction' order. Because every message carries a sequence number and the hash of the previous one, inside the signed payload, the next message exposes the hole immediately. WhatsApp or Signal wouldn't surface this; for a hospital or a command chain, it's the difference that matters."

13. Export signed evidence and verify it independently — ~90 sec

12.1 In either GUI window, click **Export Evidence** (top bar, left of the fingerprint strip). A notice appears in the Conversation panel: `Evidence exported: exports/evidence_<you>_<peer>_<time>.json`, with the exact verify command underneath. (The file is gitignored.)

12.2 Run the scripted version, which also demonstrates tampering:

```
python demo/run_evidence_demo.py
```

12.3 Expected output: the verifier report for the genuine file ends with

```
RESULT: VALID ✓ every signature, chain link and the export signature checked out.
```

then, after the demo edits one word (`Approve` → `Reject`) in record #1 of a copy:

```
✗ #1 ...: signature -- RSA signature does NOT verify (message or chain fields were altered, or wrong key)
✗ #1 ...: record hash -- record_hash does not match the record's content
✗ #3 ...: chain -- prev_hash does not link to the previous record: ...
✗ exporter signature by ... is INVALID -- the file was edited after export ...
RESULT: INVALID ✗ 4 check(s) failed; first failure: record #1 -- signature.
...
DEMO 5 PASSED
```

To verify a GUI export yourself instead (optional): `python tools/verify_transcript.py exports/<file>.json` . To pin a key you trust, add `--expect-fingerprint "XXXX XXXX XXXX XXXX"` (one of the fingerprints from step 5).

Say: "This is the opposite of WhatsApp's deniability. Every message is signed by its author, so this file proves who said what, and the verifier is a separate program that imports none of our chat code, so an auditor doesn't have to trust us, either user, or the server. Change one word anywhere and it names the exact message and the exact check that failed."

14. Closing statement — ~30 sec

Say: "Going back to our Review 1 problem statement: we set out to build a chat system where the relay server — which has to exist, to actually deliver messages — is never a point where message content, passwords, or private keys are exposed, and where standard network-level attacks — eavesdropping, tampering, replay, and man-in-the-middle — are all defended against, not just encrypted-and-hoped. And beyond what WhatsApp or Signal set out to do, it's an auditable messenger: every message is attributable to its author, a relay silently dropping a message is detected, and a signed evidence file can be verified by a third party. And beyond even that, the system is self-monitoring: an Attack Lab that triggers real relay attacks on demand, a login lockout that stops password guessing, and a live security dashboard tying it all together. What you've just seen is that working end to end: a real GUI with a security receipt on every message, a real wire log showing exactly what crosses the network, live attacks triggered from inside the app and caught by specific, testable defenses, and a dashboard watching all of it — all backed by 337 automated tests in our repo. Happy to answer questions or run any of this again."

Timing summary

Step	Approx. time
2. Launch two GUIs	0:30
3. Register alice	0:45
4. Register bob	0:45
5. Compare fingerprints	0:45
6. Exchange messages	1:30
7. Attack Lab (tamper/drop/replay/MITM)	3:30
8. Wireshark (optional)	1:00
9-12. Backup standalone scripts (optional/skip)	4:15
13. Evidence export + verifier	1:30

Step	Approx. time
14. Closing	0:30
Core demo (1-7, 13, 14, no backups)	~9:30
Core demo + Wireshark (step 8)	~10:30
Full walkthrough, including all backup scripts	~14:45